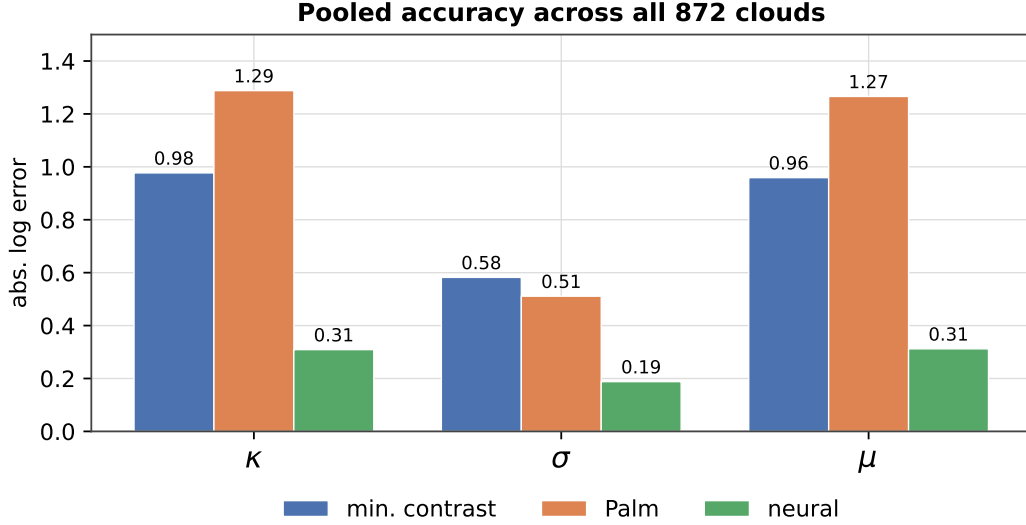
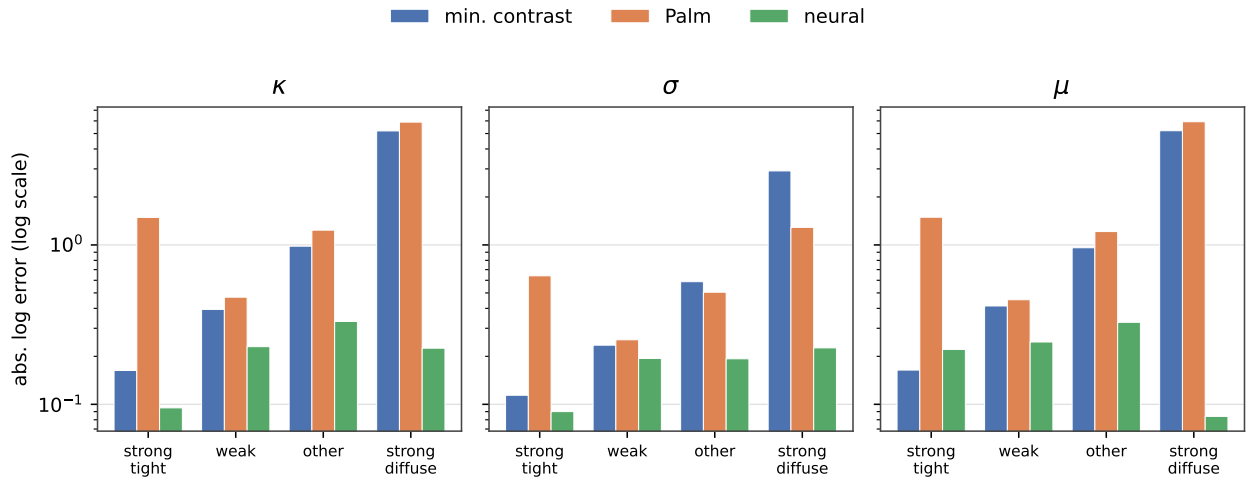


Thomas Process Estimators: Results Summary

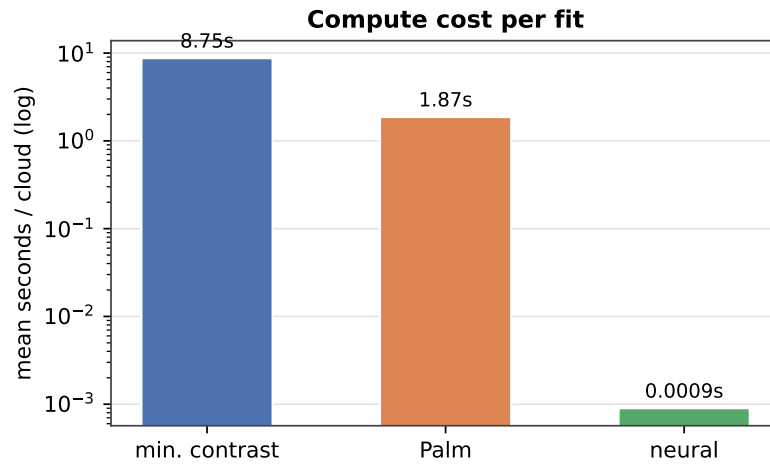
- Three estimators for (κ, σ, μ) : **min. contrast** & **Palm** (classical, training-free) vs. **neural** (persistent-homology NN).
- Held-out benchmark: 872 adversarial clouds, stratified into regimes.



- Neural lowest log-error on every parameter, pooled.
- Classical methods comparable; min. contrast edges Palm on κ and μ .



- Neural wins every regime–parameter cell *except* μ in **strong_tight** (min. contrast wins there).
- **strong_diffuse**: both classical methods collapse (logMAE $\approx 5-6$); Palm defaults to a near-Poisson fit, min. contrast blows up on a flat K -function.



- Neural $\sim 10^4 \times$ faster than either classical method; 0 failures (mincontrast/palm: 1 each).
- Palm $\sim 5 \times$ faster than min. contrast but least accurate overall.
- **Caveat:** results are in-distribution; neural sees the full cloud, classical methods a ~ 900 -pt sub-window.